Styles, Configuration, and Legacy Code

Revision: `4405d167fc323f33be76e2c7f6614fd21cc574a8`. This volume covers the files that shape packaging, deployment, presentation, documentation and backward compatibility. Appendix A contains exact text source for all four CSS files, project-level configuration, and legacy generator code.

1. CSS loading and cascade

The dashboard template loads `dashboard.css`, then `metro-theme.css`, then `dashboard-polish.css`. Browser cascade considers origin, importance, specificity and source order; among otherwise equal declarations, later rules win. This means `dashboard-polish.css` is an override layer, not a separate component runtime. `result.html` uses the same three stylesheets. `landing.html` uses landing styling and the shared theme styles.

`dashboard.css` (427 lines)

The base dashboard stylesheet defines shared color/spacing tokens, body/font defaults, header/navigation, cards, forms and controls, badges, KPI grid, charts, tables, explanation blocks, loading/status states, and responsive breakpoints. Read selectors as contracts with HTML class names: for example `.dashboard-container` sets the main content width, `.card` supplies panel surfaces/borders/shadows, `.kpi-grid` lays out key metrics, and `.chart-container` gives Chart.js a sizing context. CSS variables let later theme rules replace surface/text/border colors without duplicating every component selector.

Responsive `@media` blocks change grid/column behavior at narrower widths. If a dashboard element overflows on mobile, inspect both the base layout declaration and later breakpoint override. CSS does not decide demand, API state or schedule values.

`metro-theme.css` (121 lines)

This stylesheet defines light/dark custom properties keyed by `data-theme` on the root element, then applies them to surfaces, text, borders, controls, charts and theme-toggle affordances. The JavaScript writes `document.documentElement.dataset.theme`; CSS reacts to that attribute. Theme keys should be consumed through `var(--token)` for consistent switching. Hard-coded colors in templates/JS may not follow the theme and are potential consistency issues.

`dashboard-polish.css` (306 lines)

This file refines visual hierarchy, spacing, desktop widths, station summary, filters, tables and responsive behavior after base rules. Since it loads later, same-specificity overrides take precedence. When a declaration appears ineffective, inspect computed styles and selector specificity rather than assuming the browser ignored it. The file is not a component framework: its rules rely on the actual HTML selectors.

`landing.css` (121 lines)

Landing-only sections include hero layout, preview chart, feature cards, workflow, transparency content, and scroll-linked metro illustration. Class names also serve as hooks for `landing.js`; the `.scroll-track`, `.scroll-train` and `.scroll-track-progress` elements are positioned/animated from scroll progress. Reveal classes coordinate initial hidden/visible states with `IntersectionObserver`. A `prefers-reduced-motion` rule should keep the content available without requiring animation. The JS additionally honors reduced motion for train travel.

2. Runtime and deployment configuration

`run.py` and `Procfile`

`run.py` imports the Flask app and, only when run as the main script, reads `PORT` (default 5000), derives debug mode from a small accepted set of strings, and starts Flask on `0.0.0.0`. The Render `Procfile` says `web: gunicorn run:app`: Gunicorn imports module `run` and serves the exported WSGI object named `app`. In production, the `if __name__ == '__main__'` development-server block is not the serving mechanism.

Importing `run` imports `app`, routes, forecasting and graph dependencies. Module-level model/CSV loads therefore occur during worker startup; a missing incompatible model or required cleaned data can prevent service startup. `requirements.txt` declares Python packages but is not a lockfile. It uses a mix of exact pins and lower bounds; a future compatible version may be installed for minimum-bound entries. SciPy is declared but repository source imports were not found. `gunicorn` was appended without a version pin.

`.gitignore`

It excludes `venv/`, Python bytecode/cache and the two generated, large training-data CSV paths. This controls Git tracking only. It does not delete local files and does not prevent pipeline scripts from writing those paths. Runtime inference loads the saved joblib model/metadata and station CSV; training pipeline 03 reads `data/synthetic/dmrc_calibrated_footfall.csv`. A deployment that needs only serving does not necessarily need the large training CSV, but retraining needs it or a way to regenerate it.

`requirements.txt`

One requirement per line describes installed dependencies. Flask/Jinja/Werkzeug serve HTTP/templates; pandas/NumPy handle tables and arrays; scikit-learn supplies estimator classes; joblib serializes them; python-dateutil/date helpers serve date calculations; Gunicorn runs WSGI. Compatibility risk: joblib pickle-like artifacts depend on compatible Python/library versions and trusted artifacts. There is no lockfile, container image, or CI environment tracked.

`.vscode/launch.json` and `.vscode/settings.json`

These are editor conveniences, not app settings. The launch profile runs `run.py` in VS Code's integrated terminal with `FLASK_DEBUG=1`, Jinja debugging and debugpy. The settings choose `${workspaceFolder}/venv/bin/python` and terminal activation. Render does not read these editor files.

3. Documentation and data provenance

`read.md` describes user setup and project usage; treat implementation and a fresh code audit as authoritative where prose conflicts. `DATA_AUDIT.md` records data provenance/limits. The raw station file is the seed for cleaning; operations CSV contains scenario assumptions; line split JSON contains estimates; generated training CSVs are ignored; model binaries are produced by pipeline 03; metrics JSON records synthetic holdout evaluation. These are different data roles and should not be conflated.

Missing from tracked files: no automated test directory/configuration, no CI workflow, no Dockerfile, no database schema/migration, and no dependency lockfile. Do not present any as project features. Lack of tests does not prove behavior is correct; it means regressions need manual checks unless tests are added.

4. Legacy generators

`csvFilesGenerator/footfall.py` is an older generator, not the numbered production pipeline. It reads the compatibility station CSV, calculates station weights from hard-coded line and hub weights, produces 12 weeks of hourly values using seeded Poisson sampling, and writes the ignored legacy CSV. It differs from pipeline 02 in timespan, model factors and metadata outputs. The shared output filename can mislead operators; verify which script generated a CSV before describing it.

`csvFilesGenerator/compartmentcapacity.py` defines a small table of line/coach/capacity values and writes `metro_lines.csv` in the current working directory. Its output location depends on where the shell runs it, unlike pipeline paths anchored to `BASE_DIR`. It is a legacy utility and does not feed the current scheduler, which reads `data/cleaned/metro_line_operations.csv`.

`csvFilesGenerator/howtorun.txt` is a human run note. It is not executable configuration. Its final command lacks a newline between `python run.py` and the next prompt in the tracked text; this is a documentation formatting issue, not Python syntax.

5. Static assets and generated artifacts

`app/static/animation.gif` and `app/static/dmrc_logo.png` are tracked image files, not line-addressable source. Repository search found no template/CSS/JS references to these two assets at the documented revision, so they appear unused by the current UI. (An untracked external consumer cannot be ruled out.) The `.joblib` model and metadata are binary serialized artifacts; pipeline 03 produces them and `app/forecasting.py` loads them. `evaluation_metrics.json` is text and is read for the evaluation endpoint/display. Model files should only be loaded from trusted project-controlled sources.

6. Reliability, security and targeted improvements

- Check every module-level file read and model load in a clean deployment image; import-time exceptions can block all routes.
- Validate operations data (positive headways, capacities, fleet, turnaround) before calculations.
- Pin a tested dependency set and ensure model artifacts were created with compatible versions.
- Replace dynamic `innerHTML` in browser renderers with DOM nodes plus `textContent` for station/line/search strings.
- Add route tests for malformed JSON, unknown station, invalid time/date, missing model files, closed service and graph failures.

• Add offline pipeline tests with tiny fixture data; verify deterministic seed behavior and output schema.

• Clarify that "estimated"/"synthetic" values are scenario outputs rather than live counts or observed ridership.

Appendix

Complete text sources follow with original line numbers for CSS, deployment/editor config, dependency files, docs, and legacy utilities. Image/model/data binaries are catalogued by path, format, producer/consumer and limitations in the Master Index rather than printed as text.

Appendix A - complete source excerpts

These excerpts come from the documented commit. Original line numbers are printed at left. For print readability, Unicode dash variants are normalized to ASCII hyphens and other non-ASCII source symbols are shown as explicit Unicode escape notation; this does not change the source line numbers or code meaning. Long lines wrap with indented continuations.

Complete base dashboard stylesheet `app/static/css/dashboard.css:1-427`

```
/* DMRC Intelligent Transit Dashboard Styling */
:root {
  --dmrc-blue: #003366;
  --dmrc-red: #c8102e;
  --dmrc-dark: #1a1e24;
  --dmrc-light-bg: #f4f7f9;
  --card-bg: #ffffff;
  --border-color: #e2e8f0;
  --text-primary: #1e293b;
  --text-muted: #64748b;
  --success: #10b981;
  --warning: #f59e0b;
  --danger: #ef4444;
  --info: #0284c7;
  --font-sans: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: var(--font-sans);
  background-color: var(--dmrc-light-bg);
  color: var(--text-primary);
  line-height: 1.5;
  padding-bottom: 50px;
}

/* Header */
.dmrc-header {
  background: linear-gradient(135deg, #002244 0%, #004488 100%);
  color: white;
  padding: 18px 30px;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
}

.header-content {
  max-width: 1400px;
  margin: 0 auto;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 15px;
}

.brand-wrapper {
  display: flex;
  align-items: center;
  gap: 16px;
}

.dmrc-logo {
  height: 50px;
  width: auto;
  background: white;
  padding: 4px;
  border-radius: 6px;
}

.brand-text h1 {
  font-size: 1.4rem;
  font-weight: 700;
  letter-spacing: -0.5px;
}

.brand-text p {
  font-size: 0.85rem;
  opacity: 0.85;
}

.status-pill {
  background: rgba(255, 255, 255, 0.15);
  border: 1px solid rgba(255, 255, 255, 0.3);
  padding: 6px 14px;
  border-radius: 20px;
  font-size: 0.8rem;
  display: flex;
  align-items: center;
  gap: 8px;
}

.live-dot {
  width: 8px;
  height: 8px;
  background-color: #10b981;
  border-radius: 50%;
  animation: pulse 2s infinite;
}

@keyframes pulse {
  0% { transform: scale(0.95); box-shadow: 0 0 0 0 rgba(16, 185, 129, 0.7); }
  70% { transform: scale(1); box-shadow: 0 0 0 6px rgba(16, 185, 129, 0); }
  100% { transform: scale(0.95); box-shadow: 0 0 0 0 rgba(16, 185, 129, 0); }
}

/* Container */
.dashboard-container {
  max-width: 1400px;
  margin: 25px auto 0;
  padding: 0 20px;
}

/* Data Provenance Banner */
.provenance-banner {
  background: white;
  border: 1px solid var(--border-color);
  border-radius: 8px;
  padding: 12px 20px;
  margin-bottom: 25px;
  display: flex;
  justify-content: space-between;
  align-items: center;
```

```
  flex-wrap: wrap;
  gap: 10px;
  font-size: 0.82rem;
}

.provenance-tags {
  display: flex;
  gap: 12px;
  flex-wrap: wrap;
}

.badge {
  display: inline-flex;
  align-items: center;
  gap: 5px;
  padding: 4px 10px;
  border-radius: 6px;
  font-size: 0.75rem;
  font-weight: 600;
  letter-spacing: 0.3px;
}

.badge-verified {
  background: #dcfce7;
  color: #15803d;
  border: 1px solid #bbf7d0;
}

.badge-estimated {
  background: #fef3c7;
  color: #b45309;
  border: 1px solid #fde68a;
}

.badge-simulated {
  background: #e0f2fe;
  color: #0369a1;
  border: 1px solid #bae6fd;
}

/* Layout Grid */
.dashboard-grid {
  display: grid;
  grid-template-columns: 360px 1fr;
  gap: 25px;
}

@media (max-width: 1024px) {
  .dashboard-grid {
    grid-template-columns: 1fr;
  }
}

/* Cards */
.card {
  background: var(--card-bg);
  border: 1px solid var(--border-color);
  border-radius: 10px;
  padding: 22px;
  box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);
  margin-bottom: 25px;
}

.card-title {
  font-size: 1.05rem;
  font-weight: 700;
  color: var(--text-primary);
  margin-bottom: 16px;
  display: flex;
  align-items: center;
  justify-content: space-between;
}

/* Control Panel */
.form-group {
  margin-bottom: 16px;
}

.form-group label {
  display: block;
  font-size: 0.82rem;
  font-weight: 600;
  color: var(--text-muted);
  text-transform: uppercase;
  letter-spacing: 0.5px;
  margin-bottom: 6px;
}

.form-control, .form-select {
  width: 100%;
  padding: 10px 12px;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  font-size: 0.95rem;
  background-color: white;
  transition: border-color 0.2s;
}

.form-control:focus, .form-select:focus {
  outline: none;
  border-color: #0284c7;
  box-shadow: 0 0 0 3px rgba(2, 132, 199, 0.15);
}

.quick-buttons {
  display: flex;
  flex-wrap: wrap;
  gap: 6px;
  margin-top: 6px;
}

.calendar-context {
  margin-top: 10px;
  padding: 9px 11px;
  border-radius: 7px;
  background: #eff6ff;
  color: #334155;
  font-size: 0.82rem;
  line-height: 1.5;
```

```
}

.btn-chip {
  background: #f1f5f9;
  border: 1px solid #e2e8f0;
  padding: 4px 10px;
  border-radius: 14px;
  font-size: 0.75rem;
  cursor: pointer;
  color: var(--text-primary);
  transition: all 0.2s;
}

.btn-chip:hover, .btn-chip.active {
  background: var(--dmrc-blue);
  color: white;
  border-color: var(--dmrc-blue);
}

.slider-container {
  margin-top: 10px;
}

.slider-labels {
  display: flex;
  justify-content: space-between;
  font-size: 0.75rem;
  color: var(--text-muted);
  margin-top: 4px;
}

.btn-primary {
  width: 100%;
  background: linear-gradient(135deg, #004488 0%, #002b55 100%);
  color: white;
  border: none;
  padding: 12px 18px;
  font-size: 0.95rem;
  font-weight: 600;
  border-radius: 6px;
  cursor: pointer;
  transition: opacity 0.2s, transform 0.1s;
  box-shadow: 0 2px 6px rgba(0, 51, 102, 0.2);
}

.btn-primary:hover {
  opacity: 0.95;
}

.btn-primary:active {
  transform: translateY(1px);
}

/* KPI Grid */
.kpi-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
  gap: 16px;
  margin-bottom: 25px;
}

.kpi-card {
  background: white;
  border: 1px solid var(--border-color);
  border-radius: 8px;
  padding: 16px;
  border-left: 4px solid var(--dmrc-blue);
}

.kpi-title {
  font-size: 0.78rem;
  font-weight: 600;
  color: var(--text-muted);
  text-transform: uppercase;
}

.kpi-value {
  font-size: 1.6rem;
  font-weight: 700;
  color: var(--text-primary);
  margin: 4px 0 2px;
}

.kpi-subtext {
  font-size: 0.76rem;
  color: var(--text-muted);
}

/* Chart Card */
.chart-container {
  position: relative;
  height: 320px;
  width: 100%;
}

/* Operational Explanation Box */
.explanation-box {
  background: #f8fafc;
  border: 1px solid #e2e8f0;
  border-left: 4px solid var(--info);
  border-radius: 6px;
  padding: 16px 20px;
  margin-top: 15px;
}

.explanation-title {
  font-weight: 700;
  font-size: 0.9rem;
  margin-bottom: 6px;
  color: #0369a1;
  display: flex;
  align-items: center;
  gap: 6px;
}

.explanation-body {
  font-size: 0.88rem;
  color: #334155;
  line-height: 1.6;
```

```
}

/* Table */
.data-table {
  width: 100%;
  border-collapse: collapse;
  font-size: 0.88rem;
  margin-top: 10px;
}

.data-table th {
  background: #f8fafc;
  padding: 10px 14px;
  text-align: left;
  font-weight: 600;
  color: var(--text-muted);
  border-bottom: 2px solid var(--border-color);
  font-size: 0.8rem;
  text-transform: uppercase;
}

.data-table td {
  padding: 10px 14px;
  border-bottom: 1px solid #f1f5f9;
}

.data-table tr:hover {
  background: #f8fafc;
}

/* Metro Line Badges */
.line-tag {
  display: inline-block;
  padding: 3px 8px;
  border-radius: 4px;
  font-size: 0.75rem;
  font-weight: 600;
  color: white;
}

.line-yellow { background-color: #eab308; color: #1e293b; }
.line-blue { background-color: #2563eb; }
.line-red { background-color: #dc2626; }
.line-violet { background-color: #7c3aed; }
.line-pink { background-color: #ec4899; }
.line-magenta { background-color: #a855f7; }
.line-green { background-color: #16a34a; }
.line-orange { background-color: #ea580c; }
.line-grey { background-color: #64748b; }

/* Tabs */
.tabs-header {
  display: flex;
  gap: 10px;
  border-bottom: 1px solid var(--border-color);
  margin-bottom: 15px;
}

.tab-btn {
  background: none;
  border: none;
  padding: 8px 16px;
  font-size: 0.88rem;
  font-weight: 600;
  color: var(--text-muted);
  cursor: pointer;
  border-bottom: 2px solid transparent;
}

.tab-btn.active {
  color: var(--dmrc-blue);
  border-bottom-color: var(--dmrc-blue);
}
```

Complete dashboard refinement stylesheet `app/static/css/dashboard-polish.css:1-306`

```
/* Forecast workspace: share the landing page's calm, editorial visual system. */
.dashboard-container {
  width: min(1280px, calc(100% - 56px));
  margin: 15px auto 64px;
  padding: 0;
}
.dmrc-header {
  position: relative;
  padding: 18px max(28px, calc((100vw - 1280px) / 2));
}
.dmrc-header::after {
  position: absolute;
  right: 0;
  bottom: -1px;
  left: 0;
  height: 2px;
  content: "";
  background: linear-gradient(90deg, #0b638a 0 27%, #e0b641 27% 39%, transparent 39%);
  opacity: .8;
}
.brand-text h1 { font-size: 1.2rem; font-weight: 680; }
.brand-text h1 span { margin-left: 8px; }
.brand-text p { margin-top: 2px; font-size: .76rem; }
.header-actions { gap: 12px; }
.status-pill { padding: 7px 12px; }
.dashboard-toolbar {
  width: min(1280px, calc(100% - 56px));
  margin: 22px auto 0;
  padding: 0 0 12px;
  justify-content: flex-start;
  border-bottom: 1px solid var(--line);
  color: var(--faint);
  font-size: 10px;
  font-weight: 750;
  letter-spacing: .1em;
}
.dashboard-toolbar strong { color: var(--ink); font-size: 11px; letter-spacing: 0; }
.dashboard-toolbar .toolbar-separator { color: var(--line); }
.toolbar-route { position: relative; width: 32px; height: 15px; display: inline-flex; align-items: center; justify-content: space-between; }
.toolbar-route::before { position: absolute; top: 6px; right: 2px; left: 2px; height: 2px; content: ""; background: #8cb8ca; }
.toolbar-route i { position: relative; width: 8px; height: 8px; border: 2px solid #267ca0; border-radius: 50%; background: var(--surface); }
.toolbar-route i:last-child { border-color: #c69b34; }
.dashboard-grid { grid-template-columns: minmax(280px, 326px) minmax(0, 1fr); gap: 22px; }
.controls-column { position: relative; }
.controls-column > .card:first-child { position: relative; overflow: hidden; border-top: 2px solid #267ca0; }
.controls-column > .card:first-child::after { position: absolute; top: 62px; bottom: 24px; left: 12px; width: 1px; content: ""; background: linear-gradient(var(--line), #b4d4df 60%, var(--line)); pointer-events: none; }
.controls-column > .card:first-child .card-title,
.results-column > .card .card-title { letter-spacing: -.015em; }
.card {
  border-radius: 13px;
  padding: 21px;
  box-shadow: 0 5px 22px rgba(22, 42, 56, .045);
  margin-bottom: 17px;
}
.card-title { margin-bottom: 15px; font-size: 1rem; }
.card-title > span:first-child { display: inline-flex; align-items: center; gap: 8px; }
.controls-column > .card:first-child .card-title > span:first-child::before {
  width: 7px; height: 7px; border: 2px solid #267ca0; border-radius: 50%; content: "";
  box-shadow: 0 0 0 3px color-mix(in srgb, #267ca0 13%, transparent);
}
.form-group { margin-bottom: 17px; }
.form-group label { color: var(--muted); font-size: .7rem; letter-spacing: .08em; }
.form-control, .form-select { min-height: 43px; padding: 10px 12px; border-radius: 9px; font-size: .9rem; }
.form-control:focus, .form-select:focus { border-color: var(--accent); box-shadow: 0 0 0 3px color-mix(in srgb, var(--accent) 15%, transparent); }
.quick-buttons { gap: 6px; margin-top: 8px; }
.btn-chip { padding: 6px 10px; border-radius: 99px; font-size: .74rem; }
.calendar-context { margin-top: 10px; padding: 11px 12px; border: 1px solid color-mix(in srgb, var(--accent) 12%, var(--line)); border-radius: 9px; font-size: .78rem; line-height: 1.55; }
.slider-labels { color: var(--faint); font-size: .63rem; }
.btn-primary { width: 100%; font-weight: 680; letter-spacing: .005em; }
.results-column > .card:first-child { border-left: 3px solid #267ca0; }
.results-column > .card:first-child h2 { margin-bottom: 3px; font-size: 1.45rem !important; letter-spacing: -.035em; }
.explanation-box { margin-top: 16px; border-radius: 9px; }
.kpi-grid { display: grid; grid-template-columns: repeat(5, minmax(0, 1fr)); gap: 10px; margin: 0 0 17px; }
.kpi-card { min-width: 0; padding: 15px 14px; border: 1px solid var(--line); border-left-width: 3px !important; border-radius: 10px; box-shadow: 0 3px 12px rgba(22, 42, 56, .035); }
.kpi-title { color: var(--muted); font-size: .68rem; font-weight: 650; }
.kpi-value { margin: 6px 0 3px; font-size: clamp(1.15rem, 1.55vw, 1.55rem); font-weight: 680; letter-spacing: -.04em; }
.kpi-subtext { color: var(--faint); font-size: .66rem; line-height: 1.45; }
.chart-container { height: 285px; }
.results-column .card-title .badge { max-width: 52%; white-space: normal; text-align: right; }
.data-table { border-collapse: separate; border-spacing: 0; }
.data-table th { font-size: .68rem; letter-spacing: .045em; text-transform: uppercase; }
.data-table td { font-size: .78rem; }
.results-column > .card { position: relative; }
.results-column > .card:not(:first-child) .card-title::before {
  width: 3px; height: 18px; margin-right: 8px; border-radius: 3px; content: ""; background: #5295b1;
}
#interchangeSection .card-title,
.results-column > .card:not(:first-child) .card-title { justify-content: flex-start; }
#interchangeSection .card-title > span:first-child { margin-right: 0; text-align: left; }
.travel-card-title .travel-title-group { justify-content: flex-start; margin: 0 auto 0 0; text-align: left; }
.travel-card-title .travel-header-meta { margin-left: auto; }
html[data-theme="dark"] .dmrc-header::after { opacity: .95; }
html[data-theme="dark"] .kpi-card { box-shadow: none; }

body:not(.forecast-ready) .dashboard-grid { grid-template-columns: minmax(0, 870px); justify-content: center; }
body:not(.forecast-ready) .controls-column > .card:first-child { width: 100%; margin-bottom: 0; padding: 27px 30px; border-top: 2px solid #0066b3; }
body:not(.forecast-ready) #predictionForm { display: grid; grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 19px 24px; align-items: start; }
body:not(.forecast-ready) #predictionForm .form-group { margin-bottom: 0; }
body:not(.forecast-ready) #predictionForm > .btn-primary,
body:not(.forecast-ready) #predictionForm > .forecast-form-status { grid-column: 1 / -1; }
.forecast-context-line { margin-top: 7px; color: var(--muted); font-size: .76rem; line-height: 1.5; }
.forecast-form-status { min-height: 0; margin: -9px 0 0; color: var(--muted); font-size: .76rem; }
.forecast-form-status:empty { display: none; }
.forecast-form-status.is-error { color: #b42318; }
.section-icon { width: 17px; height: 17px; flex: 0 0 17px; fill: none; stroke: currentColor; stroke-width: 1.5; stroke-linecap: round; stroke-linejoin: round; }
.explanation-title > span { display: inline-flex; align-items: center; gap: 8px; }
.travel-count { color: var(--faint); font-weight: 500; letter-spacing: 0; }
```

```
.travel-card-title { gap: 0; }
.travel-title-group { display: flex; flex-wrap: wrap; align-items: baseline; gap: 0 6px; }
.travel-title-group > span { color: var(--ink); }
.travel-title-group > strong { color: var(--accent-strong); }
.travel-header-meta { display: flex; flex-wrap: wrap; align-items: center; justify-content: flex-end; gap: 8px; }
.network-time-badge { display: inline-flex; align-items: center; min-height: 27px; padding: 5px 9px; border: 1px solid color-mix(in srgb, #0066b3 23%, var(--line)); border-radius: 7px; color: #0066b3; background: color-mix(in srgb, #0066b3 7%, var(--surface)); font-size: 9px; font-weight: 700; letter-spacing: .035em; white-space: nowrap; }
.travel-search-wrap { margin: 0 0 13px; }
.travel-search-field { position: relative; display: block; }
.travel-search-field > svg { position: absolute; z-index: 1; top: 50%; left: 13px; width: 17px; height: 17px; transform: translateY(-50%); fill: none; stroke: var(--faint); stroke-width: 1.7; stroke-linecap: round; }
.travel-search-field input { padding-left: 40px; }
.travel-table-wrap { max-height: 430px; overflow: auto; border: 1px solid var(--line); border-radius: 10px; scrollbar-color: #b7c8d3 transparent; }
.travel-table { width: 100%; margin: 0; border-collapse: separate; border-spacing: 0; }
.travel-table thead { position: sticky; z-index: 2; top: 0; }
.travel-table th { height: 42px; padding: 10px 14px; border-bottom: 1px solid var(--line); background: var(--surface-soft); color: var(--faint); font-size: 9px; letter-spacing: .1em; }
.travel-table th:first-child, .travel-table td:first-child { width: 42%; }
.travel-table th:nth-child(2), .travel-table td:nth-child(2) { width: 31%; }
.travel-table th:last-child, .travel-table td:last-child { width: 27%; text-align: right; }
.travel-table td { height: 48px; padding: 8px 14px; border-bottom: 1px solid var(--line); color: var(--muted); font-size: .78rem; }
.travel-table tbody tr:last-child td { border-bottom: 0; }
.travel-table tbody tr:nth-child(even) { background: var(--surface-soft); }
.travel-table tbody tr:hover { background: var(--accent-soft); }
.travel-table td:first-child strong { color: var(--ink); font-weight: 620; }
.line-chip { position: relative; display: inline-grid; grid-template-columns: 7px minmax(0, 1fr); align-items: center; gap: 8px; width: 154px; min-height: 29px; padding: 4px 9px; overflow: hidden; border: 1px solid color-mix(in srgb, var(--metro-line-color) 24%, var(--line)); border-radius: 7px; color: var(--ink); background: color-mix(in srgb, var(--metro-line-color) 8%, var(--surface)); font-size: .7rem; font-weight: 650; }
.line-chip::before { position: absolute; top: 0; bottom: 0; left: 0; width: 3px; content: ""; background: var(--metro-line-color); }
.line-marker { width: 7px; height: 7px; border: 2px solid var(--metro-line-color); border-radius: 50%; background: var(--surface); }
.line-chip > span { overflow: hidden; text-overflow: ellipsis; white-space: nowrap; }
.travel-time-value { color: var(--accent-strong); font-size: .82rem; font-weight: 700; font-variant-numeric: tabular-nums; }
.travel-time-value small { color: var(--faint); font-size: .68rem; font-weight: 500; }
html[data-theme="dark"] .network-time-badge { color: #8ecde4; border-color: #345c70; background: #1d3441; }

@media (max-width: 680px) {
  body:not(.forecast-ready) #predictionForm { grid-template-columns: 1fr; gap: 17px; }
  body:not(.forecast-ready) #predictionForm > .btn-primary,
  body:not(.forecast-ready) #predictionForm > .forecast-form-status { grid-column: 1; }
  .travel-table { min-width: 570px; }
  .travel-table-wrap { max-height: 390px; }
  .line-chip { width: 142px; }
}
@media (max-width: 420px) {
  .line-chip { width: 128px; }
}

/* Keep wide analytical tables inside their own scroll areas on narrow screens. */
@media (max-width: 820px) {
  .dashboard-grid { grid-template-columns: minmax(0, 1fr); }
  body:not(.forecast-ready) .dashboard-grid,
  body.forecast-ready .dashboard-grid { grid-template-columns: minmax(0, 1fr); }
  .controls-column, .results-column, .results-column > .card { min-width: 0; }
  #interchangeSection > div[style*="grid-template-columns"] > div:last-child { min-width: 0; overflow-x: auto; }
  #interchangeSection .data-table { min-width: 590px; }
  .results-column > .card .card-title { flex-wrap: wrap; }
}
@media (max-width: 600px) {
  .dashboard-container, .dashboard-toolbar { width: calc(100% - 30px); }
  .controls-column { display: block; }
  .results-column .card-title .badge { max-width: 100%; text-align: left; }
  .results-column > .card .card-title { align-items: flex-start; }
  #interchangeSection > div[style*="grid-template-columns"] { grid-template-columns: minmax(0, 1fr) !important; }
  .travel-card-title { display: flex; flex-wrap: wrap; align-items: center; justify-content: flex-start; column-gap: 0; row-gap: 7px; }
  .travel-header-meta { justify-content: flex-start; }
  .travel-table { min-width: 0; table-layout: fixed; }
  .travel-table th:first-child, .travel-table td:first-child { width: 37%; }
  .travel-table th:nth-child(2), .travel-table td:nth-child(2) { width: 36%; }
  .travel-table th:last-child, .travel-table td:last-child { width: 27%; }
  .travel-table th { padding: 9px 6px; font-size: 7px; letter-spacing: .04em; }
  .travel-table td { padding: 8px 6px; font-size: .67rem; }
  .travel-table td:first-child strong { overflow-wrap: anywhere; }
  .line-chip { width: 100%; min-width: 0; grid-template-columns: 7px minmax(0, 1fr); gap: 5px; padding: 4px 5px; font-size: .62rem; }
  .travel-time-value { font-size: .72rem; white-space: nowrap; }
  .travel-time-value small { font-size: .59rem; }
}

/* Keep the initial scenario controls in one horizontal planning band on desktop. */
@media (min-width: 1101px) {
  body:not(.forecast-ready) .dashboard-grid {
    grid-template-columns: minmax(0, 1fr);
    justify-content: stretch;
  }
  body:not(.forecast-ready) .controls-column > .card:first-child {
    width: 100%;
    padding: 25px 30px 22px;
  }
  body:not(.forecast-ready) #predictionForm {
    grid-template-columns: minmax(190px, 1.08fr) minmax(270px, 1.35fr) minmax(220px, 1.08fr) minmax(210px, 1fr);
    gap: 18px 22px;
  }
  body:not(.forecast-ready) #predictionForm > .form-group { min-width: 0; }
  body:not(.forecast-ready) #predictionForm > .btn-primary,
  body:not(.forecast-ready) #predictionForm > .forecast-form-status { grid-column: 1 / -1; }
  body:not(.forecast-ready) .slider-labels { font-size: .62rem; }
}

/* Scenario controls stay compact above live analysis at every viewport size. */
body .dashboard-grid,
body:not(.forecast-ready) .dashboard-grid,
body.forecast-ready .dashboard-grid { grid-template-columns: minmax(0, 1fr); gap: 12px; }
body .controls-column { min-width: 0; }
body .controls-column > .card:first-child { margin-bottom: 0; padding: 16px 20px; }
body .controls-column > .card:first-child::after { display: none; }
body:not(.forecast-ready) #predictionForm,
body.forecast-ready #predictionForm { display: grid; grid-template-columns: repeat(4, minmax(0, 1fr)); gap: 12px 16px; align-items: start; }
body #predictionForm .form-group { min-width: 0; margin-bottom: 0; }
body #predictionForm .form-group label { min-height: 18px; }
body #predictionForm > .calendar-context { grid-column: 1 / -1; align-self: stretch; display: flex; align-items: center; width: 100%; min-height: 56px; box-sizing: border-box; margin: 0; padding: 11px 15px; font-size: .74rem; line-height: 1.45; }
```

```
body #predictionForm .quick-buttons { gap: 4px; }
body #predictionForm .btn-chip { padding: 5px 8px; font-size: .68rem; }
body #predictionForm .slider-labels { font-size: .57rem; }
body #predictionForm > .forecast-form-status { grid-column: 1 / -1; margin: 0; }
body .results-column { display: block; min-width: 0; animation: none; }
body .dashboard-container { width: min(1440px, calc(100% - 40px)); margin-top: 10px; }

@media (max-width: 900px) {
  body:not(.forecast-ready) #predictionForm,
  body.forecast-ready #predictionForm { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 12px; }
  body:not(.forecast-ready) #predictionForm .form-group:nth-child(2),
  body.forecast-ready #predictionForm .form-group:nth-child(2) { grid-column: 1 / -1; }
  body:not(.forecast-ready) #predictionForm > .forecast-form-status,
  body.forecast-ready #predictionForm > .forecast-form-status { grid-column: 1 / -1; }
}
@media (max-width: 600px) {
  body .dashboard-container { width: calc(100% - 24px); margin-top: 8px; }
  body .dashboard-grid { gap: 9px; }
  body .controls-column > .card:first-child { padding: 14px; }
  body:not(.forecast-ready) #predictionForm,
  body.forecast-ready #predictionForm { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 10px; }
  body:not(.forecast-ready) #predictionForm .form-group:nth-child(2),
  body.forecast-ready #predictionForm .form-group:nth-child(2) { grid-column: 1 / -1; }
  body #predictionForm > .calendar-context { min-height: 54px; padding: 10px 12px; }
  body .results-column > .card, body .kpi-grid { margin-bottom: 10px; }
  body .card { padding: 14px; }
}

@media (max-width: 1100px) {
  .dashboard-grid { grid-template-columns: 290px minmax(0, 1fr); gap: 16px; }
  .kpi-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); }
  .dashboard-container, .dashboard-toolbar { width: calc(100% - 40px); }
}
@media (max-width: 820px) {
  .dashboard-grid { grid-template-columns: 1fr; }
  .controls-column { display: grid; grid-template-columns: minmax(0, 1.25fr) minmax(220px, .75fr); gap: 14px; align-items: start;
}
  .controls-column > .card { margin-bottom: 0; }
  .controls-column > .card:first-child { grid-row: span 2; }
  .kpi-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); }
}
@media (max-width: 600px) {
  .dashboard-container, .dashboard-toolbar { width: calc(100% - 30px); }
  .dashboard-container { margin-top: 12px; }
  .dashboard-toolbar { margin-top: 17px; font-size: 9px; }
  .dmrc-header { padding: 14px 16px; }
  .brand-text h1 span { display: block; margin: 1px 0 0; font-size: .72rem; }
  .brand-text p { font-size: .67rem; }
  .header-actions { width: auto; }
  .header-actions .header-home-link { padding-inline: 6px; font-size: .75rem; }
  .theme-toggle { min-width: 42px; min-height: 42px; padding-inline: 9px; }
  .theme-toggle .theme-icon { width: 23px; height: 23px; font-size: 21px; }
  .header-actions .theme-toggle-label { display: none; }
  .status-pill { padding: 6px 9px; font-size: .7rem; }
  .controls-column { display: block; }
  .controls-column > .card { margin-bottom: 14px; }
  .card { padding: 17px; border-radius: 12px; }
  .controls-column > .card:first-child::after { left: 9px; }
  .kpi-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 8px; }
  .kpi-card { padding: 12px 11px; }
  .chart-container { height: 240px; }
  .results-column > .card .card-title { align-items: flex-start; gap: 9px; }
  .results-column .card-title .badge { max-width: 48%; font-size: .62rem; }
  #interchangeSection > div[style*="grid-template-columns"] { grid-template-columns: 1fr !important; }
  #interchangeSection > div[style*="grid-template-columns"] > div:first-child { width: min(230px, 100%); margin: auto; }
}

/* Final responsive overrides follow the legacy dashboard breakpoints above. */
@media (max-width: 820px) {
  .dashboard-grid,
  body:not(.forecast-ready) .dashboard-grid,
  body.forecast-ready .dashboard-grid { grid-template-columns: minmax(0, 1fr); }
  .controls-column, .results-column, .results-column > .card { min-width: 0; }
  #interchangeSection > div[style*="grid-template-columns"] > div:last-child { min-width: 0; overflow-x: auto; }
  #interchangeSection .data-table { min-width: 590px; }
  .results-column > .card .card-title { flex-wrap: wrap; }
}
@media (max-width: 600px) {
  .dashboard-container, .dashboard-toolbar { width: calc(100% - 30px); }
  .controls-column { display: block; }
  #interchangeSection > div[style*="grid-template-columns"] { grid-template-columns: minmax(0, 1fr) !important; }
  .results-column > .card .card-title { align-items: flex-start; }
  .results-column .card-title .badge { max-width: 100%; text-align: left; }
  .travel-card-title { display: flex; flex-wrap: wrap; align-items: center; justify-content: flex-start; column-gap: 0; row-gap:
7px; }
  .travel-card-title .network-time-badge { justify-self: start; }
  .travel-table { min-width: 0; table-layout: fixed; }
  .travel-table th:first-child, .travel-table td:first-child { width: 37%; }
  .travel-table th:nth-child(2), .travel-table td:nth-child(2) { width: 36%; }
  .travel-table th:last-child, .travel-table td:last-child { width: 27%; }
  .travel-table th { padding: 9px 6px; font-size: 7px; letter-spacing: .04em; }
  .travel-table td { padding: 8px 6px; font-size: .67rem; }
  .travel-table td:first-child strong { overflow-wrap: anywhere; }
  .line-chip { width: 100%; min-width: 0; grid-template-columns: 7px minmax(0, 1fr); gap: 5px; padding: 4px 5px; font-size:
.62rem; }
  .travel-time-value { font-size: .72rem; white-space: nowrap; }
  .travel-time-value small { font-size: .59rem; }
}
```

Complete landing stylesheet `app/static/css/landing.css:1-121`

```
* { box-sizing: border-box; }
html { scroll-behavior: smooth; }
body.landing-page { margin: 0; padding: 0; overflow-x: clip; font-family: Inter, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif; background: var(--canvas); color: var(--ink); }
.landing-page a { color: inherit; text-decoration: none; }
.site-header { position: sticky; top: 0; z-index: 30; border-bottom: 1px solid var(--line); }
.site-header-inner { height: 72px; max-width: 1240px; padding: 0 32px; margin: auto; display: flex; align-items: center; gap: 44px; }
.brand { display: inline-flex; align-items: center; gap: 10px; flex-shrink: 0; }
.brand-mark { display: inline-grid; place-items: center; width: 38px; height: 38px; border-radius: 12px; background: var(--accent-strong); color: var(--canvas); }
.brand-mark svg { width: 27px; height: 27px; fill: none; stroke: currentColor; stroke-width: 2; stroke-linecap: round; stroke-linejoin: round; }
.brand > span:last-child { display: grid; gap: 1px; }
.brand strong { color: var(--ink); font-size: 15px; letter-spacing: -.025em; }
.brand small { color: var(--muted); font-size: 10px; }
.site-nav { display: flex; align-items: center; gap: 30px; margin-left: auto; }
.site-nav a, .mobile-nav a { color: var(--muted); font-size: 13px; transition: color .2s ease; }
.site-nav a:hover, .mobile-nav a:hover { color: var(--accent-strong); }
.site-actions { display: flex; align-items: center; gap: 10px; }
.button { min-height: 40px; padding: 0 16px; display: inline-flex; align-items: center; justify-content: center; gap: 10px; border: 1px solid transparent; border-radius: 9px; font-size: 13px; font-weight: 650; transition: transform .18s ease, background .18s ease, border-color .18s ease; }
.button:hover { transform: translateY(-1px); }
.button-primary { background: var(--accent-strong); color: var(--canvas) !important; box-shadow: 0 4px 10px color-mix(in srgb, var(--accent-strong) 18%, transparent); }
.button-primary:hover { filter: brightness(1.08); }
.button-quiet { color: var(--ink) !important; background: transparent; border-color: var(--line); }
.button-quiet:hover { background: var(--surface-soft); }
.button-large { min-height: 48px; padding: 0 20px; font-size: 14px; }
.mobile-menu-toggle { display: none; width: 38px; height: 38px; color: var(--ink); background: var(--surface); border: 1px solid var(--line); border-radius: 9px; font-size: 18px; }
.mobile-nav { padding: 14px 24px 20px; border-top: 1px solid var(--line); display: grid; gap: 16px; }
.mobile-nav[hidden] { display: none; }
.section-wrap { width: min(1120px, calc(100% - 64px)); margin-inline: auto; }
.hero { position: relative; min-height: 600px; padding: 80px 0 92px; display: grid; grid-template-columns: .93fr 1.07fr; gap: 60px; align-items: center; }
.hero-copy { position: relative; z-index: 1; padding: 4px 0 20px; }
.eyebrow, .section-kicker { color: var(--accent-strong); font-size: 10px; font-weight: 760; letter-spacing: .13em; text-transform: uppercase; }
.eyebrow { display: inline-flex; align-items: center; gap: 8px; padding: 8px 11px; border: 1px solid var(--line); border-radius: 99px; background: var(--surface); letter-spacing: .08em; }
.eyebrow-dot { width: 7px; height: 7px; border-radius: 50%; background: #43a889; box-shadow: 0 0 0 3px color-mix(in srgb, #43a889 15%, transparent); }
.hero h1 { margin: 22px 0 18px; max-width: 570px; color: var(--ink); font-size: clamp(42px, 5vw, 62px); font-weight: 650; letter-spacing: -.052em; line-height: 1.02; }
.hero h1 span { display: block; color: var(--muted); font-weight: 490; }
.hero-lede { max-width: 525px; color: var(--muted); font-size: 16px; line-height: 1.75; }
.hero-actions { margin-top: 27px; display: flex; align-items: center; flex-wrap: wrap; gap: 22px; }
.text-link { display: inline-flex; align-items: center; gap: 8px; color: var(--accent-strong) !important; font-size: 13px; font-weight: 650; }
.text-link:hover { text-decoration: underline; text-underline-offset: 4px; }
.hero-proof { margin-top: 28px; max-width: 440px; display: flex; gap: 9px; align-items: flex-start; color: var(--muted); font-size: 11px; line-height: 1.55; }
.proof-icon { width: 17px; height: 17px; flex: 0 0 17px; display: inline-grid; place-items: center; border: 1px solid var(--line); border-radius: 50%; font-size: 11px; font-weight: 700; }
.preview-shell { position: relative; z-index: 2; max-width: 570px; width: 100%; justify-self: end; overflow: hidden; border: 1px solid var(--line); border-radius: 16px; background: var(--surface); box-shadow: 0 25px 70px rgba(18, 39, 55, .11), 0 3px 9px rgba(18, 39, 55, .04); transform: perspective(1200px) rotateY(-2deg) rotateX(1deg); }
.preview-topline { height: 42px; padding: 0 16px; display: flex; align-items: center; gap: 12px; color: var(--faint); font-size: 9px; font-weight: 700; letter-spacing: .1em; border-bottom: 1px solid var(--line); background: var(--surface-soft); }
.window-dots { display: flex; gap: 4px; }.window-dots i { width: 6px; height: 6px; border-radius: 50%; background: var(--line); }
.preview-status { margin-left: auto; display: inline-flex; align-items: center; gap: 6px; color: var(--muted); font-size: 9px; font-weight: 550; letter-spacing: 0; }.preview-status b { width: 6px; height: 6px; border-radius: 50%; background: #ca963e; }
.preview-content { padding: 23px 24px 20px; }
.preview-heading { display: flex; align-items: flex-start; justify-content: space-between; gap: 14px; }
.micro-label { color: var(--faint); font-size: 9px; font-weight: 750; letter-spacing: .12em; }
.preview-heading h2 { margin: 5px 0 3px; color: var(--ink); font-size: 20px; letter-spacing: -.025em; }
.preview-heading p { margin: 0; color: var(--muted); font-size: 11px; }.preview-heading p span { margin: 0 4px; }
.interchange-chip { padding: 6px 9px; color: var(--accent-strong); background: var(--accent-soft); border-radius: 6px; font-size: 9px; font-weight: 650; }
.preview-stat-row { display: grid; grid-template-columns: 1fr 1fr; gap: 10px; margin-top: 20px; }
.preview-stat { min-height: 95px; padding: 14px; display: flex; flex-direction: column; justify-content: center; border: 1px solid var(--line); border-radius: 10px; background: var(--surface); }
.preview-stat span { color: var(--muted); font-size: 10px; }
.preview-stat strong { margin-top: 5px; color: var(--ink); font-size: 23px; font-weight: 650; letter-spacing: -.04em; }
.preview-stat strong small { color: var(--muted); font-size: 10px; font-weight: 500; letter-spacing: 0; }
.preview-stat em { margin-top: 3px; color: var(--faint); font-size: 9px; font-style: normal; }
.preview-chart-heading { margin: 20px 0 8px; display: flex; justify-content: space-between; color: var(--muted); font-size: 10px; font-weight: 600; }.preview-chart-heading span:last-child { color: var(--faint); font-size: 9px; font-weight: 500; }
.preview-chart-wrap { position: relative; height: 116px; }
.preview-chart { display: block; width: 100%; height: 95px; overflow: visible; }
.chart-area { fill: url(#areaFill); }.chart-line { fill: none; stroke: #2376a8; stroke-width: 2.2; vector-effect: non-scaling-stroke; }.chart-cursor { stroke: var(--accent); stroke-width: 1; stroke-dasharray: 3 4; opacity: .55; }.chart-current { fill: var(--surface); stroke: #2376a8; stroke-width: 2.5; }
.chart-xlabels { display: flex; justify-content: space-between; color: var(--faint); font-size: 9px; }
.preview-route { margin-top: 6px; padding: 13px 12px 10px; border: 1px solid var(--line); border-radius: 9px; background: var(--surface-soft); }
.route-label { display: flex; align-items: center; gap: 7px; color: var(--muted); font-size: 8px; font-weight: 750; letter-spacing: .09em; }.route-dot { width: 6px; height: 6px; border-radius: 50%; background: #2883b0; }.route-line-name { margin-left: auto; color: var(--faint); font-size: 8px; font-weight: 550; letter-spacing: 0; }
.route-track { position: relative; height: 14px; margin: 12px 12px 5px; border-top: 2px solid #bfdae5; }.route-track-fill { position: absolute; top: -2px; left: 0; width: 53%; border-top: 2px solid #3289b1; }
.route-stop { position: absolute; top: -5px; width: 8px; height: 8px; border: 2px solid #3289b1; border-radius: 50%; background: var(--surface); }.stop-a { left: 0; }.stop-b { left: 53%; }.stop-c { right: 0; border-color: #c59a38; }
.route-train, .workflow-train, .scroll-train { position: absolute; z-index: 2; width: 32px; height: 14px; border-radius: 6px 6px 4px 4px; background: #0d638b; border: 1px solid rgba(255,255,255,.6); box-shadow: 0 2px 5px rgba(13, 67, 94, .23); }
.route-train { left: 50%; top: -9px; }.route-train i, .workflow-train i, .scroll-train i { position: absolute; top: 3px; width: 6px; height: 4px; background: #d9edf5; border-radius: 1px; }.route-train i:first-child, .workflow-train i:first-child, .scroll-train i:first-child { left: 5px; }.route-train i:nth-child(2), .workflow-train i:nth-child(2), .scroll-train i:nth-child(2) { left: 14px; }.route-train b, .workflow-train b, .scroll-train b { position: absolute; right: 3px; top: 5px; width: 3px; height: 3px; border-radius: 50%; background: #ffe59a; }
.route-stations { display: flex; justify-content: space-between; color: var(--faint); font-size: 8px; }
.preview-footnote { padding: 9px 24px; border-top: 1px solid var(--line); color: var(--faint); background: var(--surface-soft); font-size: 9px; }
.hero-background-route { position: absolute; right: -80px; bottom: 0; width: 58%; height: 100%; opacity: .12; pointer-events: none; }
.hero-background-route::before, .hero-background-route::after { content: ''; position: absolute; width: 140%; height: 40px; border: 1px solid #4c91aa; border-radius: 50%; transform: rotate(-15deg); }.hero-background-route::before { top: 40%; left: -10%; }.hero-background-route::after { top: 53%; left: 5%; }
.hero-background-route span { position: absolute; width: 8px; height: 8px; border: 2px solid #4c91aa; background: var(--canvas); border-radius: 50%; }.hero-background-route span:nth-child(1) { top: 35%; left: 38%; }.hero-background-route span:nth-child(2) { top: 54%; left: 70%; }.hero-background-route span:nth-child(3) { top: 69%; left: 50%; }.hero-background-route span:nth-child(4) { top: 27%; left: 85%; }
.trust-strip { border-top: 1px solid var(--line); border-bottom: 1px solid var(--line); background: var(--surface); }
.trust-inner { width: min(1120px, calc(100% - 64px)); min-height: 70px; margin: auto; display: flex; align-items: center;
```

```
justify-content: space-between; gap: 14px; color: var(--muted); font-size: 11px; }.trust-lead { color: var(--faint); font-size: 9px; font-weight: 750; letter-spacing: .12em; white-space: nowrap; }.trust-feature { display: inline-flex; align-items: center; gap: 7px; padding: 7px 9px; border: 1px solid var(--line); border-radius: 99px; background: var(--surface-soft); white-space: nowrap; }.trust-feature i { width: 7px; height: 7px; flex: 0 0 7px; border-radius: 50%; background: currentColor; }.trust-blue { color: #277ba6; }.trust-yellow { color: #ac8115; }.trust-green { color: #23805c; }.trust-violet { color: #7954a4; }
.platform-section { padding-top: 104px; padding-bottom: 112px; }
.section-heading { max-width: 640px; margin-bottom: 38px; }.section-heading h2, .transparency-card h2, .closing-copy h2 { margin: 13px 0 12px; color: var(--ink); font-size: clamp(30px, 3.5vw, 42px); font-weight: 620; letter-spacing: -.045em; line-height: 1.12; }.section-heading > p, .closing-copy > p { max-width: 520px; color: var(--muted); font-size: 14px; line-height: 1.7; }
.feature-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 16px; }
.feature-card { position: relative; min-height: 245px; padding: 24px 23px 27px; overflow: hidden; border: 1px solid var(--line); border-radius: 12px; background: var(--surface); box-shadow: 0 5px 20px rgba(20, 42, 58, .025); transition: transform .2s ease, box-shadow .2s ease; }.feature-card:hover { transform: translateY(-3px); box-shadow: var(--shadow-soft); }
.feature-icon { display: inline-grid; place-items: center; width: 36px; height: 36px; margin-bottom: 19px; border-radius: 10px; color: var(--accent-strong); background: var(--accent-soft); font-size: 20px; }.feature-index { display: block; color: var(--faint); font-size: 9px; font-weight: 750; letter-spacing: .1em; }.feature-card h3 { margin: 8px 0 7px; color: var(--ink); font-size: 16px; letter-spacing: -.02em; }.feature-card p { max-width: 310px; color: var(--muted); font-size: 12px; line-height: 1.65; }.feature-line { position: absolute; bottom: 0; left: 23px; width: 38px; height: 2px; border-radius: 3px; }.line-blue { background: #277ba6; }.line-yellow { background: #d2a52c; }.line-red { background: #c75658; }
.workflow-section { overflow: hidden; border-top: 1px solid var(--line); border-bottom: 1px solid var(--line); background: var(--surface-soft); }.workflow-inner { position: relative; padding-top: 94px; padding-bottom: 110px; }.workflow-inner .section-heading { margin-bottom: 34px; }.workflow-steps { position: relative; z-index: 1; display: grid; grid-template-columns: repeat(3, 1fr); gap: 40px; padding: 0; margin: 0; list-style: none; }.workflow-steps li { display: flex; align-items: flex-start; gap: 14px; }.workflow-steps li > span { flex: 0 0 32px; width: 32px; height: 32px; display: grid; place-items: center; color: var(--accent-strong); background: var(--accent-soft); border: 1px solid var(--line); border-radius: 50%; font-size: 12px; font-weight: 700; }.workflow-steps h3 { margin: 3px 0 6px; color: var(--ink); font-size: 14px; }.workflow-steps p { max-width: 260px; color: var(--muted); font-size: 12px; line-height: 1.6; }
.workflow-track { position: absolute; bottom: 41px; left: 0; width: 100%; height: 2px; background: var(--line); }.workflow-track-line { position: absolute; width: 66%; height: 2px; background: #5aa0bb; }.workflow-train { top: -7px; left: 62%; }.workflow-station { position: absolute; top: -4px; width: 8px; height: 8px; border: 2px solid #5aa0bb; border-radius: 50%; background: var(--surface); }.workflow-station.one { left: 0; }.workflow-station.two { left: 50%; }.workflow-station.three { right: 0; border-color: #c59a38; }
.transparency-section { padding-top: 90px; padding-bottom: 100px; }.transparency-card { position: relative; display: grid; grid-template-columns: 42px 1fr auto; gap: 20px; align-items: start; padding: 30px 32px; border: 1px solid var(--line); border-radius: 14px; background: var(--surface); box-shadow: var(--shadow-soft); }.transparency-icon { width: 38px; height: 38px; display: grid; place-items: center; border-radius: 11px; color: #9b6b20; background: #fbf3e4; font-size: 23px; }.transparency-card h2 { margin: 9px 0 11px; font-size: 27px; }.transparency-card p { max-width: 710px; margin-bottom: 15px; color: var(--muted); font-size: 12px; line-height: 1.7; }.transparency-badge { align-self: start; display: inline-flex; gap: 7px; align-items: center; padding: 7px 10px; color: #92651e; background: #fbf3e4; border-radius: 7px; font-size: 9px; font-weight: 650; white-space: nowrap; }.transparency-badge span { width: 6px; height: 6px; border-radius: 50%; background: #c38c2a; }
.closing-cta { min-height: 315px; padding: 25px 0 84px; display: flex; align-items: center; justify-content: space-between; gap: 50px; }.closing-copy h2 { font-size: 36px; }.closing-copy > p { margin-bottom: 20px; }.closing-map { width: min(380px, 40%); opacity: .9; }.closing-map svg { display: block; width: 100%; overflow: visible; }.map-line { fill: none; stroke-width: 5; stroke-linecap: round; }.map-blue { stroke: #4b94b5; }.map-yellow { stroke: #d0a52d; }.closing-map circle { fill: var(--surface); stroke: #4b94b5; stroke-width: 3; }.closing-map circle:nth-of-type(3), .closing-map circle:nth-of-type(4) { stroke: #d0a52d; }.map-train { fill: #0d638b; }.train-window { fill: #d9edf5; }
.site-footer { border-top: 1px solid var(--line); }.footer-inner { width: min(1120px, calc(100% - 64px)); min-height: 84px; margin: auto; display: flex; align-items: center; gap: 24px; }.footer-inner p { margin: 0 auto 0 16px; color: var(--faint); font-size: 10px; }.footer-inner > a:last-child { color: var(--muted); font-size: 11px; }
.scroll-metro { position: fixed; z-index: 15; top: 17vh; right: max(12px, calc((100vw - 1280px) / 2)); height: 66vh; width: 18px; pointer-events: none; opacity: .72; }.scroll-track { position: relative; display: block; width: 2px; height: 100%; margin: auto; background: var(--line); }.scroll-track-progress { position: absolute; inset: 0 auto auto 0; width: 2px; height: 0; background: #4b94b5; }.scroll-station { position: absolute; left: -3px; width: 8px; height: 8px; border: 2px solid #4b94b5; border-radius: 50%; background: var(--canvas); }.scroll-station.s1 { top: 0; }.scroll-station.s2 { top: 33%; }.scroll-station.s3 { top: 66%; }.scroll-station.s4 { bottom: 0; border-color: #d0a52d; }.scroll-train { left: -15px; top: 0; transition: top .12s linear; transform: scale(.78); transform-origin: center; }
.scroll-train { left: -6px; width: 14px; height: 30px; border-radius: 6px 6px 4px 4px; transform: none; box-shadow: 0 3px 10px rgba(13, 67, 94, .3); }
.scroll-train i, .scroll-train i:first-child, .scroll-train i:nth-child(2) { left: 3px; width: 6px; height: 4px; }
.scroll-train i:first-child { top: 5px; }.scroll-train i:nth-child(2) { top: 13px; }.scroll-train b { top: auto; right: 5px; bottom: 3px; }
.landing-data-notes { display: flex; flex-wrap: wrap; gap: 7px; margin-top: 14px; }
.landing-data-notes span { padding: 6px 9px; border: 1px solid var(--line); border-radius: 7px; color: var(--muted); background: var(--surface-soft); font-size: 10px; font-weight: 600; }
.theme-toggle { min-height: 42px; padding-inline: 13px; }
.theme-toggle .theme-icon { display: inline-grid; width: 22px; height: 22px; flex: 0 0 22px; place-items: center; color: var(--accent-strong); }
.theme-toggle .theme-icon svg { display: block; width: 19px; height: 19px; fill: none; stroke: currentColor; stroke-width: 1.7; stroke-linecap: round; stroke-linejoin: round; }
.reveal { opacity: 1; transform: none; transition: opacity .65s ease, transform .65s cubic-bezier(.2,.65,.25,1); }
html.motion-ready .reveal { opacity: 0; transform: translateY(13px); }
html.motion-ready .reveal.is-visible { opacity: 1; transform: translateY(0); }
html[data-theme="dark"] .preview-shell, html[data-theme="dark"] .feature-card, html[data-theme="dark"] .transparency-card { box-shadow: var(--shadow-soft); }
html[data-theme="dark"] .transparency-icon, html[data-theme="dark"] .transparency-badge { color: #f2cc78; background: #3b301b; }
html[data-theme="dark"] .transparency-badge span { background: #dfb04b; }
html[data-theme="dark"] .preview-chart .chart-line { stroke: #69b8d5; }
html[data-theme="dark"] .preview-chart .chart-current { stroke: #69b8d5; }
html[data-theme="dark"] .map-train { fill: #236b8b; }
html[data-theme="dark"] .trust-blue { color: #80c3e6; } html[data-theme="dark"] .trust-yellow { color: #e2c36c; }
html[data-theme="dark"] .trust-green { color: #77cba6; } html[data-theme="dark"] .trust-violet { color: #c1a0ea; }

@media (max-width: 1000px) {
  .site-header-inner { gap: 20px; padding-inline: 24px; }.site-nav { gap: 17px; }
  .hero { grid-template-columns: 1fr; gap: 44px; padding-top: 66px; }.hero-copy { max-width: 720px; }.preview-shell { justify-self: center; max-width: 680px; }.hero h1 { max-width: 700px; }.hero-background-route { width: 100%; }
  .trust-inner { flex-wrap: wrap; justify-content: center; padding-block: 16px; }.trust-inner span:first-child { flex-basis: 100%; text-align: center; }
}
@media (max-width: 720px) {
  .site-header-inner { height: 64px; padding: 0 17px; }.site-nav, .header-signin, .header-cta { display: none; }.site-actions { margin-left: auto; }.theme-toggle { min-height: 42px; min-width: 42px; padding: 0 10px; }.theme-toggle-label { display: none; }.mobile-menu-toggle { display: inline-block; }.brand-mark { width: 35px; height: 35px; }.section-wrap, .trust-inner, .footer-inner { width: calc(100% - 38px); }
  .hero { min-height: 0; gap: 38px; padding: 56px 0 62px; }.hero h1 { font-size: clamp(39px, 11vw, 53px); }.hero-lede { font-size: 14px; }.hero-actions { align-items: flex-start; flex-direction: column; gap: 17px; }.hero-proof { margin-top: 22px; }.preview-shell { transform: none; }.preview-content { padding: 19px 16px 17px; }.preview-stat { padding: 11px; min-height: 86px; }.preview-stat strong { font-size: 20px; }.preview-footnote { padding-inline: 16px; }.route-stations { font-size: 7px; }.preview-heading h2 { font-size: 18px; }.trust-inner { gap: 12px 17px; font-size: 10px; }.trust-inner span:first-child { text-align: left; }.hero-background-route { right: 0; width: 100%; overflow: hidden; }
  .platform-section { padding-top: 72px; padding-bottom: 72px; }.section-heading h2 { font-size: 34px; }.section-heading > p { font-size: 13px; }.feature-grid { grid-template-columns: 1fr; gap: 11px; }.feature-card { min-height: 0; padding: 20px; }.feature-icon { margin-bottom: 13px; }
  .workflow-inner { padding-block: 70px 92px; }.workflow-steps { grid-template-columns: 1fr; gap: 22px; }.workflow-steps p { max-width: 440px; }.workflow-track { left: 16px; bottom: 28px; width: calc(100% - 32px); }.workflow-track-line { width: 66%; }
  .transparency-section { padding-block: 65px; }.transparency-card { grid-template-columns: 38px 1fr; gap: 13px; padding: 22px 18px; }.transparency-card h2 { font-size: 22px; }.transparency-card p { font-size: 11px; }.transparency-badge { grid-column: 2; justify-self: start; }.closing-cta { min-height: 0; padding: 10px 0 65px; }.closing-map { display: none; }.closing-copy h2 { font-size: 32px; }.footer-inner { min-height: 104px; flex-wrap: wrap; padding-block: 16px; gap: 8px; }.footer-inner p { order: 3; flex-basis: 100%; margin: 0; }.scroll-metro { display: block; right: 5px; top: 24vh; height: 40vh; opacity: .4; }
}
@media (prefers-reduced-motion: reduce) {
  html { scroll-behavior: auto; }
  *, *::before, *::after { animation-duration: .01ms !important; animation-iteration-count: 1 !important; scroll-behavior: auto !important; transition-duration: .01ms !important; }
  .reveal { opacity: 1; transform: none; }
}
```

Complete shared theme stylesheet `app/static/css/metro-theme.css:1-121`

```
:root {
  color-scheme: light;
  --dmrc-blue: var(--accent-strong);
  --dmrc-light-bg: var(--canvas);
  --card-bg: var(--surface);
  --border-color: var(--line);
  --text-primary: var(--ink);
  --text-muted: var(--muted);
  --canvas: #f4f6f8;
  --surface: #ffffff;
  --surface-soft: #f8fafb;
  --ink: #202631;
  --muted: #657184;
  --faint: #8792a1;
  --line: #e0e4e9;
  --accent: #176b91;
  --accent-strong: #155778;
  --accent-soft: #eaf2f6;
  --shadow-soft: 0 8px 26px rgba(25, 35, 48, .045);
}

html[data-theme="dark"] {
  color-scheme: dark;
  --canvas: #141618;
  --surface: #1b1e21;
  --surface-soft: #202428;
  --ink: #edf0f2;
  --muted: #a4adb7;
  --faint: #7f8994;
  --line: #30353a;
  --accent: #63a9c6;
  --accent-strong: #83c0d9;
  --accent-soft: #202f37;
  --shadow-soft: 0 10px 30px rgba(0, 0, 0, .12);
}

body { background: var(--canvas); color: var(--ink); }
.dmrc-header { background: var(--surface); color: var(--ink); border-bottom: 1px solid var(--line); box-shadow: none; }
.brand-text h1 { color: var(--ink); }
.brand-text p, .status-pill { color: var(--muted); }
.status-pill { background: var(--surface-soft); border-color: var(--line); }
.provenance-banner, .card, .kpi-card { background: var(--surface); color: var(--ink); border-color: var(--line); box-shadow: var(--shadow-soft); }
.card-title, .kpi-value { color: var(--ink); }
.form-control, .form-select { background: var(--surface); color: var(--ink); border-color: var(--line); }
.calendar-context { background: var(--accent-soft); color: var(--ink); }
.btn-chip { background: var(--surface-soft); color: var(--ink); border-color: var(--line); }
.btn-chip:hover, .btn-chip.active { background: var(--accent-strong); color: var(--canvas); border-color: var(--accent-strong); }
.btn-primary { background: var(--accent-strong); color: var(--canvas); box-shadow: none; }
.explanation-box { background: var(--surface-soft); border-color: var(--line); border-left-color: var(--accent); }
.explanation-title { color: var(--accent-strong); }
.explanation-body, .data-table td { color: var(--muted); }
.data-table th { background: var(--surface-soft); color: var(--muted); border-color: var(--line); }
.data-table td { border-color: var(--line); }
.data-table tr:hover { background: var(--surface-soft); }
.tabs-header { border-color: var(--line); }
.tab-btn.active { color: var(--accent-strong); border-bottom-color: var(--accent-strong); }
.result-header, .dashboard-toolbar { background: var(--surface); color: var(--ink); border-color: var(--line); }
.result-header h1, .result-header h2, .result-header p { color: var(--ink); }

.site-header, .site-footer, .mobile-nav { background: var(--surface); color: var(--ink); border-color: var(--line); }
.theme-toggle { color: var(--muted); border-color: var(--line); background: var(--surface-soft); }
.theme-toggle:hover { color: var(--ink); border-color: var(--accent); }

.dashboard-toolbar {
  max-width: 1400px;
  margin: 18px auto -7px;
  padding: 0 20px;
  display: flex;
  align-items: center;
  justify-content: flex-end;
  gap: 10px;
}
.dashboard-toolbar a, .dashboard-toolbar button { font: inherit; }
.dashboard-toolbar a { color: var(--muted); text-decoration: none; font-size: .86rem; padding: 7px 10px; border-radius: 8px; }
.dashboard-toolbar a:hover { color: var(--accent-strong); background: var(--accent-soft); }
.theme-toggle {
  display: inline-flex; align-items: center; gap: 8px; min-height: 36px; padding: 0 12px;
  border: 1px solid var(--line); border-radius: 9px; font: 600 .82rem var(--font-sans); cursor: pointer;
}
.theme-toggle .theme-icon { display: inline-grid; width: 20px; height: 20px; flex: 0 0 20px; place-items: center; color: var(--accent-strong); }
.theme-toggle .theme-icon svg { display: block; width: 18px; height: 18px; fill: none; stroke: currentColor; stroke-width: 1.7; stroke-linecap: round; stroke-linejoin: round; }

html[data-theme="dark"] .badge-estimated { background: #3b301b; color: #f2cc78; border-color: #695526; }
html[data-theme="dark"] .badge-simulated { background: #183544; color: #92d9ef; border-color: #2a596e; }
html[data-theme="dark"] .badge-verified { background: #183b2d; color: #86d7ae; border-color: #2a624a; }
html[data-theme="dark"] .dmrc-logo { filter: brightness(.94); }

.brand-wrapper { gap: 12px; }
.brand-mark { display: inline-grid; place-items: center; width: 42px; height: 42px; flex: 0 0 42px; color: var(--canvas); background: var(--accent-strong); border-radius: 12px; }
.brand-mark svg { width: 27px; height: 27px; fill: none; stroke: currentColor; stroke-width: 2; stroke-linecap: round; stroke-linejoin: round; }
.brand-text h1 { font-size: 1.12rem; letter-spacing: -.035em; }
.brand-text h1 span { color: var(--muted); font-size: .78rem; font-weight: 500; }
.brand-text p { font-size: .75rem; }
.header-actions { display: flex; align-items: center; justify-content: flex-end; gap: 10px; }
.header-home-link { display: inline-flex; align-items: center; justify-content: center; min-height: 36px; padding: 0 13px; border: 1px solid var(--line); border-radius: 9px; color: var(--muted); background: var(--surface-soft); font-size: .84rem; text-decoration: none; transition: color .18s ease, border-color .18s ease, background .18s ease; }
.header-home-link:hover { color: var(--accent-strong); border-color: var(--accent); background: var(--accent-soft); }
.dmrc-header { padding: 14px 30px; }
.dashboard-container { margin-top: 22px; }
.dashboard-toolbar { color: var(--faint); font-size: .76rem; }
.toolbar-separator { color: var(--line); }
.provenance-banner { border-radius: 12px; }
.card { border-radius: 13px; padding: 22px; }
.kpi-card { border-radius: 11px; }
.form-control, .form-select { border-radius: 9px; min-height: 42px; }
.btn-primary { border-radius: 9px; min-height: 44px; }
html[data-theme="dark"] [style*="background: white"],
html[data-theme="dark"] [style*="background-color: white"],
html[data-theme="dark"] [style*="background: #f8fafc"] { background: var(--surface-soft) !important; }
html[data-theme="dark"] [style*="color: #64748b"],
```

```
html[data-theme="dark"] [style*="color: #334155"],
html[data-theme="dark"] [style*="color: #003366"] { color: var(--muted) !important; }

@media (max-width: 600px) {
  .dmrc-header { padding: 12px 16px; }
  .brand-text h1 { font-size: 1rem; }
  .brand-text p { max-width: 225px; font-size: .68rem; }
  .header-actions { width: 100%; justify-content: space-between; }
  .header-content { justify-content: flex-start; }
  .dashboard-toolbar { margin: 12px auto -9px; padding: 0 14px; }
  .dashboard-toolbar .theme-toggle-label { display: none; }
}
```

Complete application entrypoint `run.py:1-7`

```
import os
from app import app

if __name__ == '__main__':
    port = int(os.environ.get('PORT', 5000))
    debug = os.environ.get('FLASK_DEBUG', '0') in ('1', 'true', 'True')
    app.run(host='0.0.0.0', port=port, debug=debug)
```

Render WSGI process command `Procfile:1-1`

```
web: gunicorn run:app
```

Complete Python dependency declaration `requirements.txt:1-16`

```
blinker==1.9.0
click==8.4.2
Flask==3.1.3
itsdangerous==2.2.0
Jinja2==3.1.6
joblib>=1.4.0
MarkupSafe==3.0.3
numpy==2.5.2
pandas==3.0.5
python-dateutil==2.9.0.post0
scikit-learn>=1.5.0
scipy>=1.10.0
six==1.17.0
Werkzeug==3.1.8

gunicorn
```

Complete Git ignore configuration `.gitignore:1-7`

```
venv/
__pycache__/
*.py[cod]

# Generated training data; regenerate with pipeline/02_generate_calibrated_dataset.py
csvFiles/dmrc_footfall_data.csv
data/synthetic/dmrc_calibrated_footfall.csv
```

VS Code launch profile `.vscode/launch.json:1-18`

```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Run DMRC Predictor (Flask)",
      "type": "debugpy",
      "request": "launch",
      "program": "${workspaceFolder}/run.py",
      "console": "integratedTerminal",
      "env": {
        "FLASK_ENV": "development",
        "FLASK_DEBUG": "1"
      },
      "jinja": true,
      "justMyCode": true
    }
  ]
}
```

VS Code workspace settings `.vscode/settings.json:1-4`

```
{
  "python.defaultInterpreterPath": "${workspaceFolder}/venv/bin/python",
  "python.terminal.activateEnvironment": true
}
```

Complete legacy 12-week synthetic generator `csvFilesGenerator/footfall.py:1-162`

```
"""
Rebuilds dmrc_footfall_data.csv as an actually-structured model instead of
uniform random noise. What was wrong with the original (see git history):

  1. Only 7 calendar dates were ever generated (one per weekday) -- so
     "footfall for a Monday" was one arbitrary random draw, not an average
     of anything.
  2. generate_footfall() only knew two things: is this one of 4 hardcoded
     station names, and which of 5 time-of-day buckets are we in. Every
     other station and every weekday were treated identically.
  3. Nothing was calibrated to any real number -- the totals could have
     been anything and looked the same.

This version instead:
  - Generates 12 full weeks (84 days), so every weekday has 12 independent
    samples the app can average over instead of trusting one draw.
  - Gives every station a demand *weight* derived from real structure: how
    many lines it interchanges between (from network_topology.py) and which
    line it's on (DMRC's busiest lines carry more riders than its quietest
    ones). A handful of stations that are well-known to be disproportionately
    busy even among interchanges (Rajiv Chowk, Kashmere Gate, New Delhi) get
    a documented extra boost on top of that.
  - Distinguishes weekday / Saturday / Sunday ridership levels and uses a
    different hour-of-day shape for weekday commute traffic vs. weekend
    leisure traffic.
  - Calibrates the whole network to Delhi Metro's own published average
    daily ridership (~2.76 million/day, per Wikipedia's "List of Delhi Metro
    stations" citing Hindustan Times) rather than an arbitrary range, and
    generates at HOURLY granularity (not per-minute) -- per-minute precision
    on synthetic data was fake precision; a station's ridership doesn't
    actually vary minute to minute, an operations planner thinks in hours.
  - Uses numpy's Poisson distribution for the per-station-hour draw, so
    variance scales naturally with the expected count instead of being an
    arbitrary randint() range.

Run: python3 footfall.py   (writes ../csvFiles/dmrc_footfall_data.csv)
"""
import os
import csv
from datetime import date, timedelta

import numpy as np

import sys
sys.path.insert(0, os.path.join(os.path.dirname(__file__), '..', 'app'))
from network_topology import INTERCHANGES  # noqa: E402

HERE = os.path.dirname(os.path.abspath(__file__))
METRO_DATA_PATH = os.path.join(HERE, '..', 'csvFiles', 'DELHI_METRO_DATA.csv')
OUTPUT_PATH = os.path.join(HERE, '..', 'csvFiles', 'dmrc_footfall_data.csv')

NUM_WEEKS = 12
START_DATE = date(2025, 1, 6)  # a Monday -- keeps full weeks aligned
SYSTEM_DAILY_RIDERSHIP_WEEKDAY = 2_760_000  # DMRC's published average, sourced above

# Relative demand per line -- approximate, illustrative ordering based on
# each line's well-known role in the network (Blue/Yellow are the oldest,
# longest, most central lines; Grey/Airport Express are the network's
# lightest-used). NOT DMRC's actual published per-line ridership breakdown,
# which isn't public at this granularity -- documented here as an assumption,
# not a fact.
LINE_WEIGHT = {
    "Blue Line": 1.30, "Blue Line branch": 0.95,
    "Yellow Line": 1.25,
    "Violet Line": 1.10,
    "Pink Line": 1.05,
    "Red Line": 1.00,
    "Magenta Line": 0.90,
    "Green Line": 0.80, "Green Line branch": 0.70,
    "Grey Line": 0.40,
    "Orange Line": 0.50,  # Airport Express -- famously runs well under capacity
}
DEFAULT_LINE_WEIGHT = 0.85

# A small number of stations are known (from DMRC's own press coverage) to be
# disproportionately busier than even a typical interchange -- documented
# multiplier, not derived from data we have.
NAMED_HUB_BOOST = {
    "Rajiv Chowk": 1.6,
    "Kashmere Gate": 1.4,
    "New Delhi": 1.3,
}

# Day-type multiplier relative to a weekday, and which hourly shape to use.
DAY_TYPE = {
    0: ("weekday", 1.00), 1: ("weekday", 1.00), 2: ("weekday", 1.00),
    3: ("weekday", 1.00), 4: ("weekday", 1.00),
    5: ("saturday", 0.85),
    6: ("sunday", 0.60),
}

# Fraction of that day-type's ridership occurring in each operating hour
# (05:30-23:30). Bimodal commute peaks on weekdays; flatter, later, and
# lighter on weekends. Normalized in code, so these don't need to sum to 1.
HOUR_SHAPE = {
    "weekday": {5: 1, 6: 2.5, 7: 4.5, 8: 7.5, 9: 8.5, 10: 7, 11: 5.5, 12: 5,
                13: 5, 14: 5, 15: 5, 16: 5.5, 17: 7.5, 18: 9, 19: 8, 20: 6,
                21: 4, 22: 2.5, 23: 1},
    "saturday": {5: 0.5, 6: 1.5, 7: 2.5, 8: 4, 9: 5.5, 10: 6.5, 11: 7, 12: 7.5,
                 13: 7.5, 14: 7.5, 15: 7, 16: 6.5, 17: 6, 18: 5.5, 19: 5,
                 20: 4, 21: 2.5, 22: 1.5, 23: 1},
    "sunday": {5: 0.5, 6: 1, 7: 2, 8: 3, 9: 4.5, 10: 6, 11: 7, 12: 7.5,
               13: 7.5, 14: 7.5, 15: 7, 16: 6.5, 17: 6, 18: 5.5, 19: 5,
               20: 3.5, 21: 2, 22: 1, 23: 0.5},
}


def load_stations(csv_path):
    rows = []
    with open(csv_path, newline="", encoding="utf-8-sig") as f:
        for row in csv.DictReader(f):
            rows.append((row["Station"].strip().rstrip("\u2020\u2021*#\u00a4").strip(), row["Line"].strip()))
    return rows


def station_weights(stations):
```

```
    """Returns {(station, line): weight}, not yet normalized."""
    weights = {}
    for station, line in stations:
        w = LINE_WEIGHT.get(line, DEFAULT_LINE_WEIGHT)
        n_lines = len(INTERCHANGES.get(station, {line}))
        if n_lines > 1:
            w *= 1 + 0.8 * (n_lines - 1)
        w *= NAMED_HUB_BOOST.get(station, 1.0)
        weights[(station, line)] = w
    return weights


def generate(num_weeks=NUM_WEEKS, seed=42):
    rng = np.random.default_rng(seed)
    stations = load_stations(METRO_DATA_PATH)
    weights = station_weights(stations)
    total_weight = sum(weights.values())

    rows = []
    for week in range(num_weeks):
        for day_offset in range(7):
            d = START_DATE + timedelta(weeks=week, days=day_offset)
            day_type, day_multiplier = DAY_TYPE[d.weekday()]
            shape = HOUR_SHAPE[day_type]
            shape_total = sum(shape.values())
            daily_system_total = SYSTEM_DAILY_RIDERSHIP_WEEKDAY * day_multiplier

            for (station, line), w in weights.items():
                station_share = w / total_weight
                station_daily_total = daily_system_total * station_share
                for hour, hour_w in shape.items():
                    expected = station_daily_total * (hour_w / shape_total)
                    footfall = int(rng.poisson(max(expected, 0.1)))
                    rows.append((d.isoformat(), hour, station, line, footfall))
    return rows


if __name__ == "__main__":
    rows = generate()
    with open(OUTPUT_PATH, "w", newline="", encoding="utf-8") as f:
        writer = csv.writer(f)
        writer.writerow(["Date", "Hour", "Station", "Line", "Footfall"])
        writer.writerows(rows)
    print(f"Wrote {len(rows):,} rows to {OUTPUT_PATH}")
    print(f"Spans {NUM_WEEKS} weeks starting {START_DATE.isoformat()} "
          f"({NUM_WEEKS} samples per weekday for averaging)")
```

Complete legacy capacity CSV writer `csvFilesGenerator/compartmentcapacity.py:1-20`

```
import csv

data = [
    ["Line", "No of Coaches", "Compartment Capacity", "Total Capacity"],
    ["Red Line", 8, 250, 8 * 250],
    ["Yellow Line", 6, 250, 6 * 250],
    ["Blue Line", 8, 310, 8 * 310],
    ["Green Line", 4, 240, 4 * 240],
    ["Violet Line", 6, 240, 6 * 240],
    ["Orange Line", 6, 326, 6 * 326],
    ["Pink Line", 6, 240, 6 * 240],
    ["Magenta Line", 6, 240, 6 * 240],
    ["Grey Line", 6, 240, 6 * 240]
]

with open('metro_lines.csv', 'w', newline='') as file:
    writer = csv.writer(file)
    writer.writerows(data)

print("CSV file 'metro_lines.csv' created successfully.")
```

Legacy run instructions text `csvFilesGenerator/howtorun.txt:1-8`

```
Open a New Terminal:
Open a new terminal window or tab.

Navigate to Your Project Directory:
Use the cd command to navigate to the directory where your project is located.
Run: cd ~/CODEEE/dmrc_footfall_predictor

Run Your Application : python run.py
```

Project usage documentation as tracked `read.md:1-143`

```
# Delhi Metro Passenger Demand Forecasting & Train Scheduling Optimization System

A simulation and dashboard prototype covering **228 station records** and **11 project-defined corridors**. It does not currently contain observed DMRC station-by-hour passenger data, verified operational schedules, or verified rolling-stock constraints. See [DATA_AUDIT.md](DATA_AUDIT.md) before interpreting any forecast or recommendation.

Rather than displaying arbitrary passenger counts or naive formulas, this system:
1. **Demonstrates station-wise demand prediction** using a Gradient Boosted Tree model trained on generated synthetic demand. Its reported scores measure agreement with that synthetic generator only, not actual DMRC footfall.
2. **Estimates line-wise passenger demand** across intersecting corridors at interchange hubs (e.g., Yellow vs. Blue Line at Rajiv Chowk).
3. **Simulates frequency and headway recommendations** using configurable project assumptions for capacities, headways, turnaround, and fleet. These values are not verified DMRC standards.
4. **Delivers explainable recommendations**, providing operations controllers with transparent reasoning behind headway adjustments, capacity utilization, and platform crowding risks.
5. **Labels forecasts and line splits as synthetic estimates**, while documenting that the present operational inputs still need source verification.

---

## Architecture & Core Features

### 1. Station-Wise Simulated Demand Forecasting
* Hourly footfall predictions across all 19 operating hours (05:00 to 23:00) with smooth minute interpolation.
* Distinct modeling of weekday bimodal commute peaks (morning rush 08:00-11:00, evening rush 17:00-20:30) vs. weekend leisure curves.
* Uses code-authored demand weights and curves. These values are not verified against station-level observed DMRC ridership; do not use them as actual counts.

### 2. Interchange Line-Wise Demand Breakdown
* For the 25 multi-line interchange hubs, models passenger distribution based on corridor passenger throughput, train rake capacity, and network centrality.
* Prominently tagged as `[ESTIMATED LINE DEMAND]` to maintain absolute operational transparency.

### 3. Realistic Train Frequency & Headway Optimization
* Computes required passenger clearance frequency based on train comfort capacity (80% load factor) and through-load factors.
* **Minimum Signaling Separation**: Clamped to CATC/CBTC safety limits ($h_{\min} = 2.0$-$2.5$ min). The system will never recommend unrealistic intervals under this safety limit.
* **Maximum Policy Headway**: Clamped to service policy limits ($h_{\max} = 8.0$-$12.0$ min off-peak) to prevent passengers waiting unrealistic durations.
* **Fleet Constraint**: Limits dispatch frequency to what the line's active rolling stock and round-trip turnaround time physically support.
* **Dispatch Practical Intervals**: Rounds headways to real operational dispatch steps (e.g., 2.3 min, 2.5 min, 2.75 min, 3.0 min, 3.5 min, 5.0 min, etc.).

### 4. Network Topology & Dijkstra Route Graph
* Line-expanded network topology graph with 254 line-nodes and 25 interchanges.
* Dijkstra multi-source shortest path calculation accounting for inter-station track travel and platform transfer penalties.

### 5. Interactive Transit Operations Dashboard
* Station selector with line indicators and interchange badges.
* Date picker defaults to today's date in Delhi, permits today and future dates, and has dynamic weekday/Saturday/Sunday shortcuts. The dashboard reports weekday/workday/holiday status, listed occasion, selected-hour demand band, and that station's predicted daily peak hour.
* Real-time interactive hour slider (05:00 to 23:00) with dynamic live updates via JSON API.
* Dynamic operating-hours demand curve (ML estimate vs. synthetic training-period baseline).
* Interchange demand distribution chart and corridor performance table.
* Real-time KPI cards: Demand, Recommended Headway, Trains/Hour, Capacity Utilization (%), and Average Passenger Wait Time.

---

## Data Provenance & Integrity Standards

| Provenance | Meaning | Assets |
| :--- | :--- | :--- |
| \U0001f7e1 **Unverified project inputs** | Hand-authored route and operations tables. | Station topology and `metro_line_operations.csv`; requires external source verification. |
| \U0001f7e0 **Model estimate on synthetic data** | ML output learned from the project's generated dataset. | Forecaster; not validated against actual passenger counts. |
| \U0001f535 **Synthetic simulation** | Generated demand scenarios, separate from observed data. | `dmrc_calibrated_footfall.csv` (728 days \u00d7 19 hours \u00d7 228 stations). |

---

## Synthetic Holdout Metrics

The active model, metadata, and evaluation report have been regenerated together from the included 728-day synthetic dataset (2025-01-06 through 2027-01-03). The 23-feature `HistGradientBoostingRegressor` achieved MAE 27.90 riders/hour, RMSE 54.82, and R\u00b2 0.9947 on the chronological holdout beginning 2026-10-01 (411,540 rows). This is a holdout against generated scenarios, not real passenger counts, and should not be interpreted as real-world accuracy. Holiday/occasion impacts are assumed scenario effects until measured from observed data. See `DATA_AUDIT.md` and `models/evaluation_metrics.json` for limitations and baseline comparisons.

---

## Project Structure

```
dmrc_footfall_predictor/
\u251c\u2500\u2500 data/
\u2502   \u251c\u2500\u2500 raw/   # Preserved original source files
\u2502   \u2502   \u251c\u2500\u2500 DELHI_METRO_DATA.csv
\u2502   \u2502   \u2514\u2500\u2500 metro_lines.csv
\u2502   \u251c\u2500\u2500 cleaned/   # Derived; not independently verified
\u2502   \u2502   \u251c\u2500\u2500 stations_cleaned.csv   # Station and topology inputs
\u2502   \u2502   \u2514\u2500\u2500 metro_line_operations.csv   # Project assumptions for operations
\u2502   \u2514\u2500\u2500 synthetic/   # Simulated scenarios and estimates
\u2502   \u251c\u2500\u2500 dmrc_calibrated_footfall.csv # 728-day synthetic hourly demand
\u2502   \u2514\u2500\u2500 line_distribution_model.json # Interchange line split weights
\u251c\u2500\u2500 pipeline/
\u2502   \u251c\u2500\u2500 01_audit_and_clean_data.py   # Cleans stations and establishes ops parameters
\u2502   \u251c\u2500\u2500 02_generate_calibrated_dataset.py # Generates calibrated dataset & line split model
\u2502   \u2514\u2500\u2500 03_train_and_evaluate_forecaster.py # ML training & chronological evaluation
\u251c\u2500\u2500 models/
\u2502   \u251c\u2500\u2500 footfall_forecaster.joblib   # Serialized trained ML model
\u2502   \u251c\u2500\u2500 forecaster_metadata.joblib   # Encoders, historical means, features
\u2502   \u2514\u2500\u2500 evaluation_metrics.json   # Validation metrics comparison report
\u251c\u2500\u2500 app/
\u2502   \u251c\u2500\u2500 __init__.py   # Flask application factory
\u2502   \u251c\u2500\u2500 routes.py   # Web routes & REST API endpoints
\u2502   \u251c\u2500\u2500 forecasting.py   # Real-time ML inference & 24h curves
\u2502   \u251c\u2500\u2500 scheduling_optimizer.py   # Transit frequency & headway optimizer
\u2502   \u251c\u2500\u2500 network_topology.py   # Line sequences & interchange definitions
\u2502   \u251c\u2500\u2500 travel_graph.py   # Dijkstra shortest-time routing graph
\u2502   \u251c\u2500\u2500 templates/
\u2502   \u2502   \u251c\u2500\u2500 landing.html   # Product landing page and forecast preview
```
```

```
\u2502   \u2502   \u251c\u2500\u2500 index.html                  # Modern interactive dispatcher dashboard
\u2502   \u2502   \u2514\u2500\u2500 result.html                 # Synchronous report view
\u2502   \u2514\u2500\u2500 static/
\u2502       \u251c\u2500\u2500 css/landing.css              # Responsive landing page and metro motifs
\u2502       \u251c\u2500\u2500 css/metro-theme.css           # Shared light/dark design tokens
\u2502       \u251c\u2500\u2500 css/dashboard.css            # Transit operations styling
\u2502       \u251c\u2500\u2500 js/landing.js                # Scroll-linked train and reveal behavior
\u2502       \u251c\u2500\u2500 js/theme.js                  # Light/dark theme toggle
\u2502       \u251c\u2500\u2500 js/dashboard.js              # Dynamic Chart.js and API client
\u2502       \u2514\u2500\u2500 dmrc_logo.png
\u251c\u2500\u2500 csvFiles/                          # Synced legacy files for backward compatibility
\u251c\u2500\u2500 run.py                             # Application entry point
\u251c\u2500\u2500 requirements.txt                   # Dependencies (Flask, scikit-learn, pandas, etc.)
\u2514\u2500\u2500 Procfile                           # Gunicorn deployment configuration
```

See `DATA_AUDIT.md` for source discovery, schema/count audit, current data gaps, and use limitations.

Running Locally

1. **Activate virtual environment**:
   ```bash
   source venv/bin/activate
   pip install -r requirements.txt
   ```

2. **Run data pipeline** (optional, pre-generated models and cleaned data are included):
   ```bash
   python pipeline/01_audit_and_clean_data.py
   python pipeline/02_generate_calibrated_dataset.py
   python pipeline/03_train_and_evaluate_forecaster.py
   ```

3. **Start the web application**:
   ```bash
   python run.py
   ```

4. **Open the landing page**:
   Navigate to `http://localhost:5000`, then choose **Open forecasting workspace** to use the dashboard at `/dashboard`.

Deployment

The application is fully compatible with **Render**, **Railway**, or containerized production hosting via the included `Procfile`:
```
web: gunicorn run:app
```

Data audit report as tracked DATA_AUDIT.md:1-40

```
# Dataset audit and readiness

Audit date: 2026-10-14. This report describes files present in this repository; it does not certify them as DMRC publications.

## Findings

| Asset | Contents and validation | What it can support | Limitations |
| --- | --- | --- | --- |
| `data/raw/DELHI_METRO_DATA.csv` (228 rows) | Station, line, latitude, longitude. No timestamps or passenger counts. | Station directory and approximate topology input. | Source, retrieval date, license, accuracy, and station coverage are undocumented. It cannot train footfall forecasts. |
| `data/raw/metro_lines.csv` (9 data rows) | Line, coach count, compartment capacity, total capacity. | Starting point for configurable capacity scenarios. | No source citation, date, occupancy definition, fleets, turnaround, signaling limits, or schedules. Not verified operational data. |
| `data/synthetic/dmrc_calibrated_footfall.csv` (3,153,696 records) | 228 stations \u00d7 728 dates \u00d7 19 hours (2025-01-06 through 2027-01-03), generated by `pipeline/02_generate_calibrated_dataset.py`. | UI demonstrations and simulation experiments. | Every passenger count is synthetic. Calibration constants, station weights, hourly curves, seasonal factors, and event multipliers are code-authored assumptions, not observed AFC counts. It is not valid evidence of real-world forecast accuracy. |
| `data/synthetic/line_distribution_model.json` | Weighted synthetic line shares for interchange stations. | Scenario allocation only. | No observed OD, AFC, transfer, or line boarding data. It cannot establish real line-wise demand. |
| `data/cleaned/stations_cleaned.csv` | Derived station map and hand-authored interchange metadata. | Network demo and station selection. | "Cleaned" does not mean independently verified; the project topology covers 228 stations and is not synchronized to the newer official static feed. |
| `data/cleaned/metro_line_operations.csv` | Hand-authored operational parameters in `pipeline/01_audit_and_clean_data.py`. | What-if scheduling under explicit assumptions. | Fleet sizes, turnaround, headways, signaling minima, load factors and capacities have no row-level traceable references. Do not describe these as official/verified. |
| `models/*` and `models/evaluation_metrics.json` | Model trained on 23 features; metrics cover 2,742,156 training and 411,540 time-held-out test records, with training through 2026-09-30 and test dates from 2026-10-01 to 2027-01-03. Training historical aggregates use prior dates only. | Measures model behavior on the generated synthetic patterns and supports repeatable simulation comparisons. | This still cannot establish real-demand accuracy. The historical aggregate features are derived from synthetic data; evaluation scores must not be generalized to real DMRC counts. |

## Data gaps

- **Station/time footfall:** no observed count series, dates, intervals, or AFC-derived station entries/exits are present. Forecasts are simulation outputs only.
- **Peak/off-peak:** the synthetic generator encodes assumed hour shapes; there is no observed data to measure actual peak patterns or seasonal effects.
- **Line-wise/interchange demand:** no OD matrices, transfer counts, or line-level passenger counts are present. Current shares are weighted assumptions.
- **Train load and capacity:** project values exist but are not traceably sourced or operationally validated. Boarding clearance is an additional model assumption.
- **Timetable and feasible dispatch:** no current trip-level timetable, service calendar, rolling-stock assignment, turnaround/depot constraints, or signaling documentation is included.
- **Network map:** a hand-authored line sequence exists. Official OTD static DMRC GTFS is publicly listed with stops, routes, stop times, and trips, but last updated Aug. 10, 2023; it is a useful schedule/topology candidate, not current real-time service or passenger counts. The portal requires an access-purpose submission to download, so it has not been silently imported.
- **Model evaluation:** a synthetic chronological split measures reproduction of the generator's own patterns, not out-of-sample performance on observed passengers. No observed-data benchmark is currently possible.

## External source discovery

- Delhi Open Transit Data, [DMRC Static Data](https://otd.delhi.gov.in/data/staticDMRC/): lists 262 stops, 36 routes, 411,686 stop-time records, and 17,997 trips, last updated Aug. 10, 2023. The portal describes stop times as approximate arrival/departure times and requires submission of usage purpose before download. Suitable candidate for static route and timetable replacement after download, license/terms review, and schema audit. Does not provide passenger counts.
- DMRC, [Annual Report 2023-24](https://backend.delhimetrorail.com/documents/7611/Annual-Report-2023-24-English.pdf): reports network-wide journeys and network scope, but this is not a station-by-hour demand dataset. It cannot validate this model's station/hour predictions.

No external records were downloaded or merged into project data in this pass. This avoids treating a portal listing or a network-wide ridership statistic as station-level ground truth.

## Use and release conditions

Use the dashboard only for simulation and software demonstration until verified observed footfall and operational inputs are supplied. Keep raw downloaded data immutable; store cleaned feeds under `data/cleaned/` and generated scenarios under `data/synthetic/`. Record source URL, retrieval date, license/terms, schema checks, and transformations for every import. Retrain and regenerate the evaluation report as one pipeline run; do not publish current accuracy metrics as real-world model performance.

## Audit status

Implemented: explicit synthetic-data/API provenance, date-aware inference for 23 features, a 104-week synthetic scenario set, and a chronological holdout evaluation. The dashboard accepts today and future dates, computes weekdays for any date, and displays listed occasions and the source/status of their calendar entry. Its occasion list is populated through 2028, while fixed annual holidays are included through 2035; an unlisted occasion is not automatically inferred. Not achieved: reliable real-world forecasting or operational timetable optimization. Those require observed passenger counts, traceable train operating data, refreshed network inputs, and validation against held-out observed periods.
```